

1. Shipping as a vector for marine IAS

List of Philippine Ports is in Appendix 3

Shipping remains as the only scientifically documented pathway for marine biological invasion in the Philippines with the introduction and invasion of the South American mussel *Mytella strigata* (Vallejo et al. 2017). This invasive was first recorded from the South Harbor of Manila in 2014 and has been known to have spread throughout Manila Bay, to Lingayen Gulf, Aparri, Cagayan and Batangas Port in the Philippines. It has since then reported in Singapore, Taiwan, Hong Kong, India, Malaysia, the Gulf of Thailand, and Sri Lanka.



Figure 2. Foulers from the South Harbor of Manila Bay.
Photo by SAILS-PORTEC Manila Bay

Mytella was likely spread through hull fouling and ballast water release. In the Philippines its spread to other ports was likely through small vessel hull fouling as the first adult samples were recorded from the fishing boat FV Ocean in 2015 which was docked in Manila Bay. An intensive monitoring of the South Harbor area in 2014 resulted in the detection of the first cohort of recruits in Manila Bay. The likely first introduction by ballast water release or by biofouling was in December 2013 and the first cohort of recruits was detected in July 2014.

There are at least 15 marine non-indigenous species ship hull fouling recorded from Manila Bay's South Harbor (Vallejo et al. 2019; Trinidad et al 2017.) Only *Mytella* is considered invasive enough to have wide scale ecological and economic impacts. The most numerous species is the well-studied *Hydroides elegans*, which is a known ship fouler with a present pantropical distribution.